

7

Geg:

$$v = 100 \frac{\text{km}}{\text{h}} \triangleq 27,78 \frac{\text{m}}{\text{s}}$$

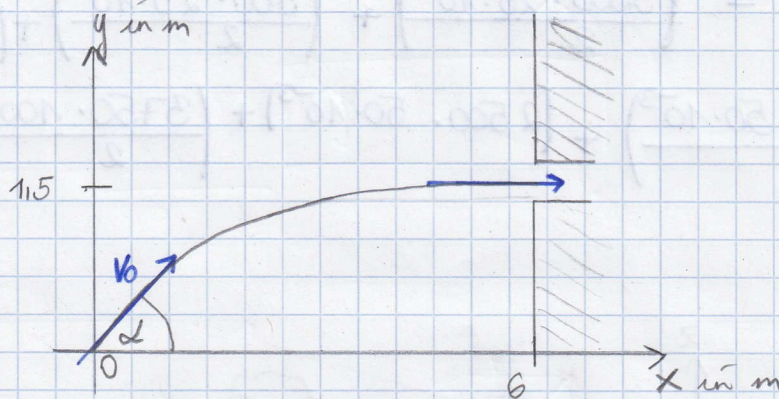
$$\Delta s = 0,75 \text{ m}$$

$$v(s) = \pm \sqrt{v_0^2 + 2a \cdot \Delta s}$$

$$0 = v_0^2 + 2a \cdot \Delta s \Rightarrow a = \frac{-v_0^2}{-2 \cdot \Delta s} = \frac{(27,78 \text{ m/s})^2}{2 \cdot 0,75 \text{ m}} = \underline{\underline{514,49 \frac{\text{m}}{\text{s}^2}}}$$

Al: Die Beschleunigung des Autos beträgt $514,49 \frac{\text{m}}{\text{s}^2}$.

8



$$x_w = 6 \text{ m}$$

$$y_h = 1,5 \text{ m}$$

$$x_w = \frac{v_0^2 \cdot \sin 2\alpha}{g} = \frac{v_0^2 \cdot 2 \sin \alpha \cos \alpha}{g}$$

$$y_h = \frac{v_0^2 \cdot \sin^2 \alpha}{2g} = \frac{v_0^2 \cdot \sin \alpha \cdot \sin \alpha}{2g}$$

$$\frac{x_w}{y_h} = \frac{\frac{v_0^2 \cdot 2 \sin \alpha \cos \alpha}{g}}{\frac{v_0^2 \cdot \sin \alpha \cdot \sin \alpha}{2g}} = \frac{6}{1,5} = \frac{2 \cdot \cos \alpha}{\sin \alpha}$$

$$4 \cdot \sin \alpha = 2 \cdot \cos \alpha$$

$$\frac{\sin \alpha}{\cos \alpha} = \frac{2}{4} \Rightarrow \tan \alpha = \frac{1}{2}$$

$$\alpha = \arctan\left(\frac{1}{2}\right) = \underline{\underline{26,57^\circ}}$$

⑧

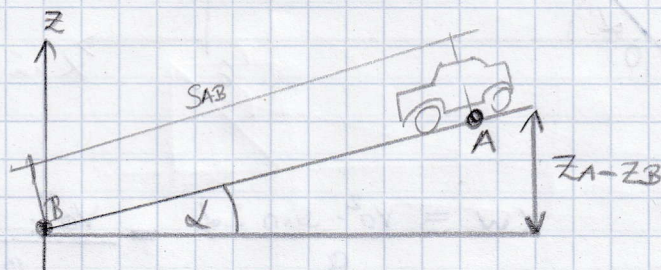
$$G = \frac{v_0^2 \cdot \sin(2 \cdot 26,5^\circ)}{g}$$

$$\sqrt{\frac{G \cdot g}{\sin(2 \cdot 26,5^\circ)}} = v_0 = \underline{\underline{8,58 \frac{m}{s}}}$$

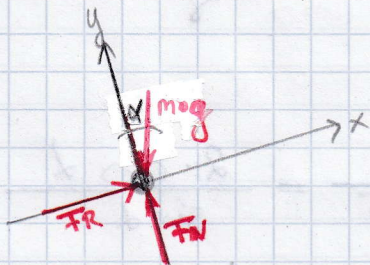
⑨

$$\begin{aligned} \vec{F} &= F \cdot t = \left(\frac{3000 \cdot 25 \cdot 10^{-3}}{2} \right) + \left(\frac{500 \cdot 25 \cdot 10^{-3}}{2} \right) + (2500 \cdot 25 \cdot 10^{-3}) + \\ &\quad \left(\frac{1250 \cdot 50 \cdot 10^{-3}}{2} \right) + (2500 \cdot 50 \cdot 10^{-3}) + \left(\frac{3750 \cdot 100 \cdot 10^{-3}}{2} \right) \\ &= \underline{\underline{450 Ns}} \end{aligned}$$

⑩

Freischaeden

geg $m = 1200 \text{ kg}$, $v = 70 \text{ km/h}$,
 $\alpha = 10^\circ$, $\mu = 0,3$



$$I: 0 = F_N - m \cdot g \cdot \cos \alpha \Rightarrow F_N = m \cdot g \cdot \cos \alpha$$

$$F_R = \mu \cdot F_N = \mu \cdot m \cdot g \cdot \cos \alpha$$

10

$$W_R = -\mu \cdot m \cdot g \cdot \cos \alpha \cdot s_{AB} \dots \text{Reibungsarbeit}$$

$$W_H = m \cdot g \cdot \sin \alpha \cdot s_{AB}$$

$$W = W_R + W_H$$

$$W = \underbrace{E_{kinB}}_0 - E_{kinA}$$

$$W = -E_{kinA}$$

$$\sin \alpha = \frac{z_A - z_B}{s_{AB}}$$

$$z_A - z_B = \sin \alpha \cdot s_{AB}$$

$$(-\mu \cdot m \cdot g \cdot \cos \alpha \cdot s_{AB}) + (m \cdot g \cdot \sin \alpha \cdot s_{AB}) =$$

$$-\frac{m \cdot v^2}{2}$$

$$s_{AB} \cdot m \cdot (-\mu \cdot g \cdot \cos \alpha + g \sin \alpha) = -\frac{m \cdot v^2}{2}$$

$$s_{AB} = \frac{-v^2}{2g(-\mu \cdot \cos \alpha + \sin \alpha)} = \frac{\left(\frac{70 \text{ km/h}}{3,6}\right)^2}{2 \cdot 9,81 \cdot (-0,3 \cdot \cos 10^\circ + \sin 10^\circ)} = \underline{\underline{158,22 \text{ m}}}$$